Copernicus Marine Service

QUALITY INFORMATION DOCUMENT

Sea Ice TAC Production Centre
Arctic Sea & Sea-Ice Surface Temperature
SEAICE_ARC_PHY_CLIMATE_L4_MY_011_016
SEAICE_ARC_PHY_CLIMATE_L3S_MY_011_021

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CHANGE RECORD

When the quality of the products changes, the QuID is updated, and a row is added to this table.

The third column specifies which sections or sub-sections have been updated. The fourth column should mention the reason for updating the document. Column 5 should include the names of the author(s) doing the changes. Including the name in column 6 shows that the document has been validated by the TAC/MFC PQ leader or deputy (or TAC/MFC PQ leader or the deputy). This validation means the document was checked for contents, formats and consistency with other documents.

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4.0	May 2025	All	New template and temporal extension	Ioanna Karagali, Wiebke M. Kolbe, Guisella Gacitúa	C. Wettre

ACRONYM TABLE

Acronym	Description
Argo	International profiling float network (www.argo.net)
AATSR	Advanced Along-Track Scanning Radiometer
ATSR	Along Track Scanning Radiometer
AVHRR	Advanced Very High Resolution Radiometer
C3S	Copernicus Climate Change Service
CCI	Climate Change Initiative
CDR	Climate Data Record
CRREL	Cold Regions Research and Engineering Laboratory
DMI	Danish Meteorological Institute
EAN	Estimated Accuracy Number
ECMWF	European Centre for Medium-Range Weather Forecasts
EiS	Entry into Service
ENVISAT	ENVIronmental SATellite
EOCIS	UK Earth Observation Climate Information Service
ERS	European Remote Sensing
ESA	European Space Agency
EUMETSAT	European Meteorological Satellites
GCOS	Global Climate Observing System
GHRSST	Group for High Resolution Sea Surface Temperature
HADIOD	Hadley Centre Integrated Ocean Database
ICDR	Interim Climate Data Record
IST	Sea-Ice Surface Temperature
MCAS	Marine and Climate Advisory Service
MetOp	Meteorological Operational Satellites
MY	Multi-year
NASA	National Aeronautics and Space Administration
NOAA	National Oceanic and Atmospheric Administration
NP	North Pole
NRT	Near Real Time
NWC SAF	Nowcasting Satellite Application Facility
OI	Optimal Interpolation
OSTIA	Operational Sea Surface Temperature and Ice Analysis
PMW	Passive Microwave
PPS	Polar Platform System
REP	Reprocessing
SAF	Satellite Application Facility
SI	Sea Ice

SIC	Sea Ice Concentration
SLSTR	Sea and Land Surface Temperature Radiometer
SST	Sea Surface Temperature
TAC	Copernicus Marine Service Thematic Assembly Centre
TIR	Thermal Infrared

KEY DEFINITIONS

2 meter air temperature	The air temperature measured at 2 meters height, T_{2m}
Sea Surface Temperature	The temperature of the ocean water close to the surface.
Sea-Ice Surface Temperature	Representative of the surface temperature over sea ice.
Thermal skin layer	The layer within which the long-wave radiation penetrates the ocean, typically less than 1mm, where ocean-atmosphere heat exchange takes place and temperature is measured using Infra-red sensors.
$SST_{sub\text{-}skin}$	Sea Surface Temperature at the base of the thermal skin layer.
SST_{skin}	Measurement of the sea surface temperature at the very top layer, or "skin," of the ocean.
$SST_{foundation}$	Sea Surface Temperature at the base of the thermal layer, which is considered representative of night-time and/or well mixed conditions, i.e, free of any diurnal signal.
Sea Ice Concentration	Fraction (percentage) of the sea surface area of the grid cell containing sea-ice.
Sea Ice	Corresponds to ocean surface covered by sea ice (>70%) thus the surface temperature is classified as sea-ice surface temperature (IST).
Level-2 Pre processed (L2P) data product	Geophysical variables derived from Level 1 source data on the Level 1 grid (typically the satellite swath projection). Ancillary data and metadata added following Group for High Resolution Sea Surface Temperature (GHRSST) Data Specification.
Level-3 Uncollated (L3U) data product	L2 data granules remapped to a regular latitude/longitude grid without combining observations from multiple source files. L3U files will typically be “sparse”, corresponding to a single satellite orbit.
Level-3 Collated (L3C) data product	SST observations from a single instrument combined into a space-time grid. In this project, a typical L3C file may contain all the observations from a single instrument in a 24-hour period.
Level-3 Super-Collated (L3S) data product	SST observations from many instruments combined into a space-time grid. In this project, a typical L3S file may contain all the observations from all available instruments in a 24-hour period.
Level 4 (L4) data product	SST observations from multiple instruments using an analysis system (e.g., optimal interpolation) to produce a gridded, gap-free product.

RELEVANT DOCUMENT LIST

CMEMS-SEAICE-SQO-011-016-021	The Synthesis Quality Overview document is a short summary of the quality information presented here. It can be found in the dedicated product page[1] under Documentation[2]
CMEMS-SEAICE-PUM-011-016-021	The Product User Manual presents the product details along with file structure. It can be found in the dedicated product page under Documentation[3]

[1]https://data.marine.copernicus.eu/product/SEAICE_ARC_PHY_CLIMATE_L4_MY_011_016/description
[2]https://documentation.marine.copernicus.eu/SQO/CMEMS-SEAICE-SQO-011-016-021.pdf
[3]https://documentation.marine.copernicus.eu/PUM/CMEMS-SI-PUM-011-016-021.pdf

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I EXECUTIVE SUMMARY

I.1 Products covered by this document

SEAICE_ARC_PHY_CLIMATE_L4_MY_011_016: The daily, gap-free, multi-year reanalysis of Ice and Sea Surface Temperature of the Arctic Ocean, for the sea surface (SST), sea ice (IST) and marginal ice zone (MIZT) temperature at 0.05 degrees resolution.

SEAICE_ARC_PHY_CLIMATE_L3S_MY_011_021: The daily super-collated L3S product of Ice and Sea Surface Temperature of the Arctic Ocean, for the sea surface (SST), sea ice (IST) and marginal ice zone (MIZT) temperature at 0.05 degrees resolution.

Both products cover the period from January 1st 1982 to December 31st 2024 and the products are extended temporally once a year, to include the period up until 1 year before present. This update depends on the availability of upstream data sources.

I.2 Summary of the results

The quality of the L4 SST and IST has been assessed using in situ observations. The SST component has been validated using drifting buoys, as well as independent mooring buoys and Argo observations from the Hadley Centre Integrated Ocean Database (HADIOD). Due to lack of available in situ skin surface temperatures over ice, validation of the IST component (T_{skin}) has been performed using air temperature measurements at 2m (T_{2m}) from Cold Regions Research and Engineering Laboratory (CRREL) ice mass balance buoys[4], European Centre for Medium-Range Weather Forecasts (ECMWF) distributed buoys[5], and 14 Russian North Pole (NP) drifting ice stations[6]. The physical difference between T_{skin} and T_{2m} introduces a difference in the comparisons, which can be several degrees and is not related to the performance of the product. The sparse in situ data coverage in the Arctic will likely not improve the metrics for the product due to the limited coverage if included in the analysis and are therefore withheld to assure independent validation statistics.

The quality of the L3S SST and IST has been assessed using the same in situ datasets as for the L4 SST and IST with the exception that the IST component is not validated against the NASA Operation IceBridge campaign data[7].

L4 SST: This variable has zero biases and standard deviation of differences less than 0.57°C when compared against drifting buoys, moored buoys and Argo floats. Validation statistics from January to May throughout the record from 1982 to May 2021 show a stable mean performance against all in situ observations: drifting buoys, moored buoys and Argo floats. The comparisons against all three types of in situ observations show a slight tendency towards smaller standard deviations in the second half of the record compared to the first half. This is probably due to more accurate input satellite products in the recent times. Note that the annual validation statistics for 2021 are not based on a full year but only cover January through May 2021. The number of drifting and moored buoy in situ observations available for validation is very low in the beginning of the record and much larger by the end of the record. This, thus, makes the yearly validation statistics less reliable in the beginning of the period.

[4] See The CRREL-Dartmouth Mass Balance Buoy Program https://imb-crrel-dartmouth.org/
[5] See ECMWF Observation DataBase https://codes.ecmwf.int/odb/conventionalkind/
[6] See AARI Russian North Polar Drifting Station Data https://gdex.ucar.edu/datasets/d474000/
[7] See NSIDC IceBridge data set pages https://nsidc.org/data/icebridge/data

L4 IST: This variable presents biases of -3.21°C, -2.87°C and -2.35°C and standard deviation of differences of 3.34°C, 3.36°C and 3.12°C, using air temperature measurements from ECMWF distributed buoys, CRREL buoys and NP drifting ice stations, respectively. The ice surface temperatures cover a limited region of the domain as the sea ice extent varies through the years. The physical difference between T_{skin} and T_{2m} introduced between both measures, can be of several degrees and not related to the performance of the product.

2024 temporal extension – L4 SST: Shows a bias of –0.06°C and standard deviation of 0.96°C against drifting buoy observations for the period June to October 2024.

2024 temporal extension – L4 IST: Shows a bias of –2.56°C and standard deviation of 3.52°C against SIMB3 buoy observations for the period June to September 2024.

L3S SST: Shows biases around 0.1°C and standard deviation of differences less than 0.54°C when compared against drifting and moored buoys and Argo floats for the period January 1982 to May 2021.

L3S IST: Shows biases of -3.22°C, -2.69°C and -1.81°C and standard deviation of differences 3.36°C, 3.34°C and 2.51°C, using air temperature measurements from ECMWF and CRREL buoys and NP drifting ice stations respectively, for the period January 1982 to May 2021.

2024 temporal extension – L3S SST: Shows a bias of –0.12°C and standard deviation of 0.94°C against drifting buoy observations for the period June to October 2024.

2024 temporal extension – L3S IST: Shows a bias of –1.40°C and standard deviation of 2.98°C against SIMB3 buoy observations for the period June to September 2024.

Important information or **noticeable event**: During the course of this reanalysis, re-processing took place in 2023 and the full time-series was replaced.

I.3 Estimated Accuracy Numbers

A summary of the estimated accuracy numbers, is presented in Table 1, derived from the validation of the L4 product and the L3S product against drifting buoys, Argo floats and moored buoys for the SST component. For the IST component, various types of sea ice buoys (ECMWF, CRREL, NP) have been used which typically record air temperature at 2 m rather than sea-ice surface skin temperature.

Table 1. Table of estimated accuracy numbers (where variables, units and numbers are given). Metrics are derived from comparison between the product value and the in-situ measurements; mean bias, standard deviation (STD), Root mean square (RMS) and Number of match-ups with in-situ observations.

Variable	**Unit**	**In Situ type**	**Metrics and "Values of metric"**			
			mean bias	STD	RMS	Number of match-ups
Sea Surface Temperature SEAICE_ARC_PHY_CLIMATE_L4_MY_011_**016** (01/01/1982-31/05-2021)	°C	Drifting buoys	0.00	0.54	0.54	3,062,549
		Moored buoys	0.03	0.56	0.56	76,052
		Argo	0.03	0.51	0.51	32,953

Variable	**Unit**	**In Situ type**	**Metrics and "Values of metric"**			
			mean bias	STD	RMS	Number of match-ups
Ice Surface Temperature SEAICE_ARC_PHY_CLIMATE_L4_MY_011_**016** (01/01/1982-31/05-2021)	°C	ECMWF buoys (T2m)	-3.21	3.34	4.63	55,288
		CRREL buoys (T2m)	-2.87	3.36	4.42	22,979
		NP Drifting Ice Stations (T2m)	-2.35	3.12	3.91	7,665
		IceBridge KT-19	1.52	3.12	3.48	36,638
Sea Surface Temperature SEAICE_ARC_PHY_CLIMATE_L3S_MY_011_**021** (01/01/1982-31/05-2021)	°C	Drifting buoys	0.07	0.43	0.44	876709
		Moored buoys	0.10	0.53	0.54	29946
		Argo floats	0.12	0.40	0.41	10312
Ice Surface Temperature SEAICE_ARC_PHY_CLIMATE_L3S_MY_011_**021** (01/01/1982-31/05-2021)	°C	ECMWF buoys (T2m)	-3.22	3.36	4.65	26502
		CRREL buoys (T2m)	-2.69	3.38	4.32	11149
		NP drifting ice stations (T2m)	-1.81	2.51	3.09	1183

II PRODUCTION SYSTEM DESCRIPTION

Production centre name: Danish Meteorological Institute, DMI, Denmark

Production system name: Arctic Sea Surface and Sea Ice Surface Temperature Reprocessed L3S and L4 Analysis (Copernicus Marine Environment Monitoring Service names: SEAICE_ARC_PHY_CLIMATE_L4_MY_011_016, SEAICE_ARC_PHY_CLIMATE_L3S_MY_011_021).

Description: The Danish Meteorological Institute Optimal Interpolation (DMIOI) Reprocessed system is run at frequent intervals for the production of the temporal extensions and is presented in Figure 1, where all input data on surface temperatures over sea, sea ice and the marginal ice zone (SST, IST, MIZT) along with the input Sea Ice Concentration (SIC) fields and the land mask are combined into the final OI (Optimal Interpolation) L4 product. More information is available in Nielsen-Englyst et al. (2023).

The final level 4 analysis product is a merged and interpolated daily field covering the Arctic, from 58° to 90° Latitude and -180° to 180° Longitude, with a 0.05° resolution in latitude and longitude and covers surface temperatures in the ocean (SST), the sea ice (IST) and the marginal ice zone (MIZ). The optimal interpolation method used to construct the merged and gap free SST/IST analysis is taken from the high latitude SST DMI processing scheme described in Høyer & She (2007), and Høyer et al. (2014).

The OI method works with anomalies from a first guess field. In the current approach, a persistence-based method is applied, which uses the previous analysis field as the first guess field. The SST and IST observations from the last 24 hours are therefore interpreted as anomalies with respect to the first guess field. The OI method will for each grid point find the solution that has the lowest errors, given statistical input, such as a first guess error variance, error covariance functions and uncertainties on the individual observations. Due to the different physical conditions for ocean and sea ice surface temperature variability, separate statistics have been derived for the open ocean and the sea ice covered regions.

The SST first guess variance and error covariance were very similar to the ones that were presented in Høyer & She (2007), and Høyer et al. (2014). Over sea ice, the first guess IST variability was derived using one year of MetOp Advanced Very High Resolution Radiometer (AVHRR) L3 aggregated observations. The previous day L4 field was subtracted from the L3 IST observations and a spatial two dimensional field of standard deviations were calculated for one year of anomalies. The first guess field and the error covariance used in the MIZ is a weighted linear combination of the open waters and the ice values, where the ice concentration was used as the weighting factor. In this context, the MIZ is defined as areas where the ice concentration is between 15% and 70%. The search radius for the OI method is set to 100 km and the maximum number of satellite observations included in the optimal estimation is 20. The average number of satellite observations included in the analysis is 18.6, indicating a good satellite data coverage within the search limits.

The product aims to be a long, consistently processed counterpart to the near real time product SEAICE_ARC_SEAICE_L4_NRT_OBSERVATIONS_011-008. For the reanalysis product SEAICE_ARC_PHY_CLIMATE_L4_MY_011_016 an additional bias correction is performed based on the SST validation against independent in situ observations using drifting buoy observations from Atkinson et al. (2014) and a clear-sky bias correction has been applied over the sea ice according to the findings in Richter-Menge et al. (2017), using the sea ice concentration as a scaling factor and a value of 0.85°C in case of 100% sea ice cover.

For the reanalysis product SEAICE_ARC_PHY_CLIMATE_L3S_MY_011_021 no additional bias correction has been applied.

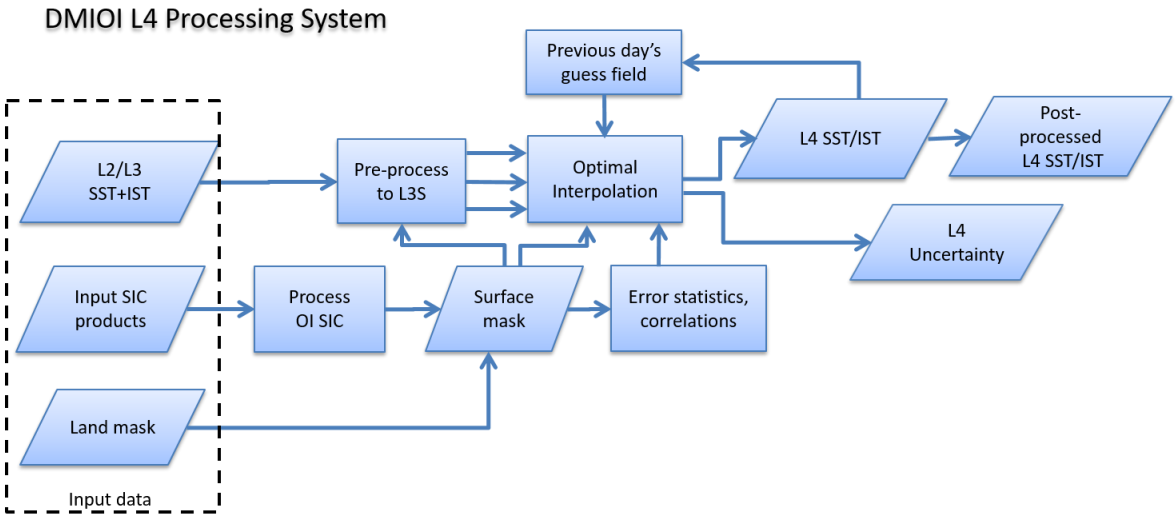

Figure 1. Schematic diagram of the operational processing chain of the Optimal Interpolation (OI) system at the Danish Meteorological Institute (DMI).

II.1 Input data

The following data are collected as input to the reprocessing system, called DMI-OI:

- **SST CCI**: The SST CCI version 2.1 data are used for the period January 1982 to December 31st 2016, obtained through the ESA SST_cci[8] project (Merchant et al., 2019). The selected fields are the sea surface temperature retrievals at 20 cm depth. The ESA CCI data series include observations from the Along Track Scanning Radiometer-1 (ATSR 1) instrument on board the European Remote Sensing-1 (ERS-1) satellite, ATSR 2 on board the ERS-2 satellite, the Advanced Along-Track Scanning Radiometer (AATSR) on board the ENVIronmental SATellite (ENVISAT), the Advanced Very High Resolution Radiometer (AVHRR) onboard the National Oceanic and Atmospheric Administration (NOAA) and Meteorological Operational (MetOp) satellites.
  Copernicus C3S SST: Use of SST observations as Level 3 Uncollated (L3U) data products available from the Copernicus Climate Change Service (C3S) Climate Data Store[9] for the period 2017-2021. The selected fields are sea surface temperature depth retrievals representative of the SST at 20 cm depth. The C3S data series include observations from the Sea and Land Surface Temperature Radiometer (SLSTR) A/B instruments on board the Sentinel 3 satellites and the AVHRRs on board MetOp-A/B satellites. For 2023-2024, the SST L3U ICDR used was funded by the UK EOCIS/MCAS. From 2025 onward, the production of the SST L3U ICDR is expected to be resumed by the C3S.
- **IST observations**: IST observations are obtained by combining a climate dataset and data from a NRT data stream: The Arctic and Antarctic ice Surface Temperatures from thermal Infrared satellite climate dataset version 2 (AASTI v2) covering the period 1982–2014 (Dybkjær et al., 2014, Kolbe et al. 2025) and data from the operational OSISAF IST product (OSI-205) covering the period from 2015 and onwards (Dybkjær et al., 2018). The AASTI and OSI-205 data are retrieved by identical algorithms from a single sensor type, the AVHRR sensors, where the applied algorithms are calibrated specifically for each sensor. The data format for the two

[8] http://cci.esa.int/
[9] https://cds.climate.copernicus.eu/

datasets differs, as AASTI uses the Global Area Coverage (GAC) and the OSI-205 uses the Local Area Coverage (LAC), at approximately 4 and 1 km resolution, respectively. For more information see Englyst-Nielsen et al. (2023).

- **Sea-ice concentration data**: The EUMETSAT OSI-SAF global sea ice concentration (SIC) climate data record v2 (1979-2015) is used together with the SICCI-25km product (2002 - May 2017) from the European Space Agency (ESA) Climate Change Initiative (CCI) programme. A special rerun of the SICCI-25km processing has been performed to provide consistent SICCI-25km SIC fields up after May 2017. The SICCI-25km product is used whenever it is available and the OSI-450 product otherwise. Land-spill over effects are more pronounced in the coarser resolution OSI-450 than in SICCI-25 km. To improve OSI-450 and to increase the consistency of the full SIC record, a number of filters have been applied to the two SIC products. Both OSI-450 and SICCI-25km have challenges in coastal regions, and for the Baltic Sea and Danish Waters high resolution SIC information is used from the Swedish Meteorological and Hydrological Institute (SMHI; 1982-2011) and 1 km SIC fields (2012-present) from the CMEMS product SEAICE_BAL_SEAICE_L4_NRT_OBSERVATIONS_011_004.

II.2 Quality control and pre-processing of input data

All satellite data valid for a particular day, within 24 hours from the analysis are considered. The input L2P SST data undergo various Quality Control (QC) and processing steps to generate separate level 3 products:

- The L3S fields have been calculated as a noise weighted average of all available observations
- Only satellite data classified as cloud free by the EUMETSAT Nowcasting Satellite Application Facility (NWC SAF) Polar Platform System (PPS) cloud mask are included.
- A minimum quality flag of 4 was used for all observations.
- Satellite observations using an ice algorithm in regions where the sea ice concentration is 0 are discarded.
- Sensor specific and quality dependent biases are subtracted as a result of the validation procedure.

The current version of the Level 2 SST and IST products do not have included dynamic uncertainties associated with the observations. The main errors in the L2 input satellite observations come from undetected clouds that enter the data and appear as cold bias in the retrievals.

III VALIDATION FRAMEWORK

The variables assessed are sea and sea-ice surface temperature from both the L4 and L3S products using the various in situ data.

III.1 Validation process for SST

The validation of SST TAC products is based on the use of drifting buoy measurements exclusively, as recommended currently by the GHRSST group on satellite SST validation (STVAL), see GHRSST (2012). Mean and standard deviation of the departures from drifting buoy SST measurements are computed over the full domain (North of 58°N). Various quality check procedures are applied by the data provider on drifting buoy measurements to discard suspect data (that result from e.g. sensor malfunction, deployment/operational issues, environmental factors, or errors during data processing and transmission), using gross error checks and/or blacklists (e.g., lists of buoys with erroneous recordings made available by various centres)[11]. Additional validation statistics were also calculated with in situ observations from moored buoys as well as Argo floats. As the different types of observations also have different error characteristics, all the validation statistics shown here will be separated into drifting buoys, moored buoys and Argo floats statistics.

The individual in situ observations have been matched to the closest satellite observations in space and time and all differences are computed as satellite – in situ. The Argo and drifter observations are well represented in the Arctic region open waters, while the moorings are concentrated in the North Atlantic and Greenland Sea.

The SST drifting buoys observations were used for a bias correction of the L4 surface temperature analysis. The moored buoys and Argo in-situ observations are not included in the reanalysis and the validation statistics and they therefore represent an independent estimate of the performance of the L4 SST analysis.

In situ SST data are used for the validation of the SST component, i.e., when sea ice concentration is less than 15%. Observations from drifting buoys, moored buoys and Argo floats are obtained from the Hadley Centre Integrated Ocean Database HADIOD version 1.2.0.0 (Atkinson et al., 2014). The spatial and temporal coverage of the three types of data are shown in Figure 2.

In addition to the overall validation statistics, results are also presented graphically for the period January 1st 1982 to May 31st 2021 for all three types of in situ observations in Figure 4 and Figure 5.

[11] See e.g. https://documentation.marine.copernicus.eu/QUID/CMEMS-INS-QUID-013-030-036.pdf

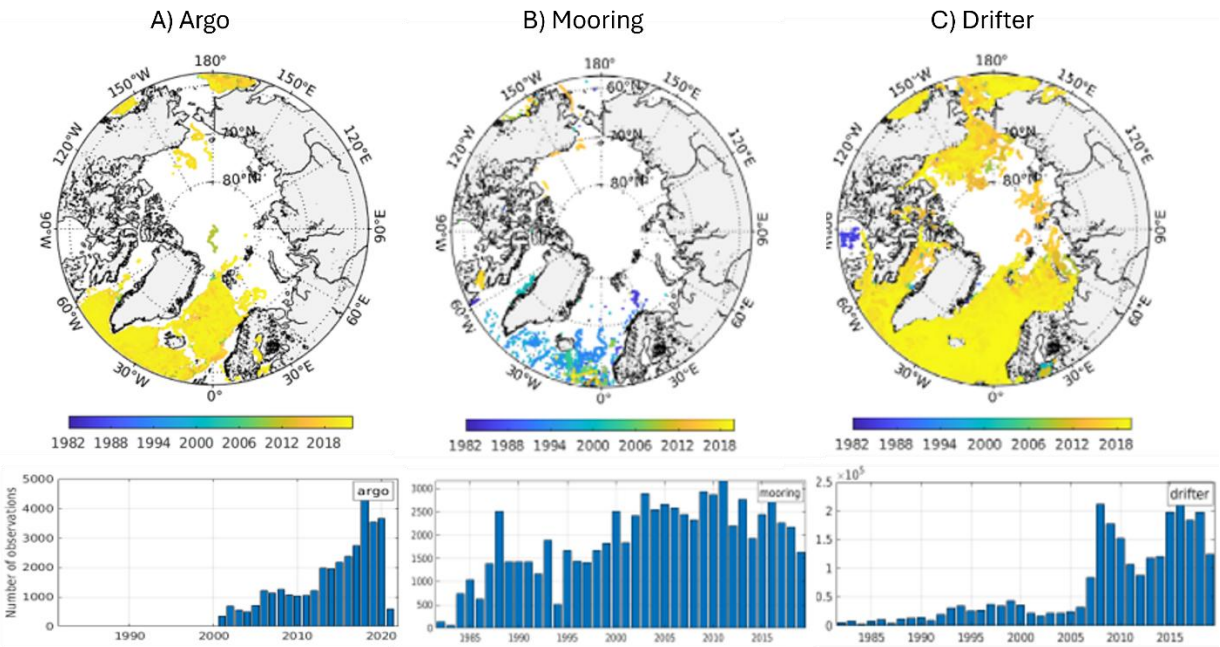

Figure 2. Overview of observations from Argo floats (A), moored buoys (B) and drifting buoys (C) used for SST validation.

III.2 Validation process for IST

The IST component has been validated against drifting buoy measurements of buoys deployed in the Arctic by ECMWF (European Centre for Medium-Range Weather Forecasts) (September 15th 1993 to January 1st 2015) and CRREL (Cold Regions Research and Engineering Laboratory) (April 14th 2001 to September 4th 2017). In addition, NP drifting ice stations were also included (1982-1991 and 2003-2012). Note that all those buoys measure the 2 m air temperature (T_{2m}), while the L4 and L3S analyses consist of surface temperatures. The physical difference between T_{skin} and T_{2m} introduces a difference between the measures, which can be of several degrees (Nielsen-Englyst et al., 2019) and not related to the performance of the product. Nonetheless, using T_{2m} measurements is the only readily available method to assess the performance of the product over ice. Finally, Ice Surface Temperature measurements from 117 NASA IceBridge flights were also used to directly validate the gap free L4 IST product. Note that due to the fact that the L3S product has gaps, which is the cause of very few match-ups with in-situ observations, the IceBridge campaign data were not used for the L3S product validation.

In situ 2-m air temperature measurements from buoys are used to validate the IST component, as sea-ice surface skin temperature is not available.

For the validation of the L4 product, in situ observations from the following independent sources was used:

- 116 drifting buoys distributed by the European Centre for Medium-Range Weather Forecasts (ECMWF), for the period September 15th 1993 to January 1st 2015,
- 96 U.S. Army Cold Regions Research Engineering Laboratory (CRREL) mass balance buoys, for the period April 14th 2001 to September 4th 2017 (Richter-Menge et al., 2017)
- 14 North Pole (NP) drifting ice stations (1982-1989, 2003-2012), see RU/FSR/HME/AARI (1993).

The temperature observations from NP, ECMWF and CRREL are air temperatures measured at different heights (typically around 2 m above the surface) depending on e.g., snow accumulation, snow drift and snow melt. These temperature observations have been inspected and quality controlled manually for data artefacts. The ECMWF and CRREL observations have previously been used for validation of satellite ISTs and corresponding 2 m air temperature estimates within the European Union's Horizon2020 project EUSTACE (EU Surface Temperatures for All Corners of Earth).

Additionally, 117 NASA IceBridge (IAKST1B) flights covering the period 2012–2019, typically conducted during March to May (version 2; Studinger 2020) was used. The surface temperatures are converted from IR radiation measurements from a Heitronics KT-19 IR Radiation Pyrometer by assuming an emissivity constant of 0.97. The surface temperatures are provided by IceBridge at a spatial resolution of about 15 m, and here they have been averaged for every fifth kilometre (mean averaged over each five kilometres) to remove the small-scale variability, which cannot be represented by the coarser resolution L4 IST product.

When lacking in-situ and or satellite data, no validation takes place.

The spatial and temporal coverage of the ECMWF and CRREL buoys are shown in Figure 3.

In addition to the overall validation statistics, results are also presented graphically for the period January 1st 1982 to May 31st 2021 for all three types of in situ observations in Figure 6 and Figure 7.

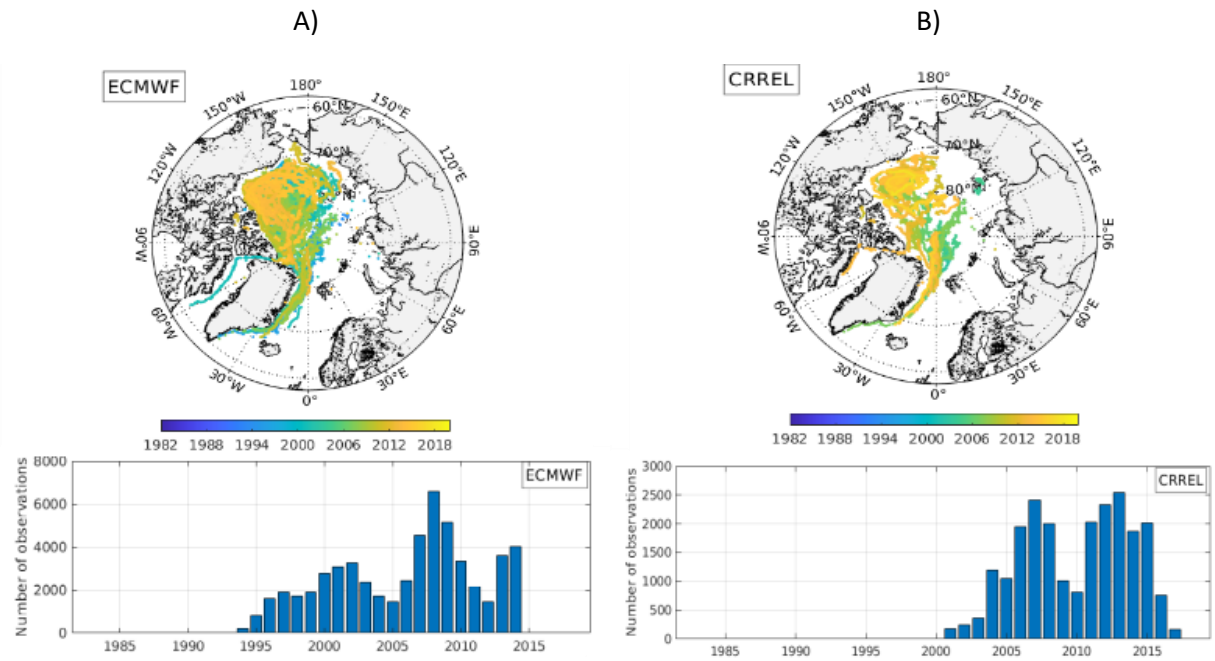

Figure 3. Overview of ECMWF (A) and CRREL (B) drifting buoys observations used for IST validation of the L3S and L4 products.

IV VALIDATION RESULTS

The following sections presents the results of the comparison of the SST and IST components of the products against in situ observations, followed by an overall conclusion on the results.

IV.1 Validation of SST

The main validation results cover the reanalysis period from January 1st 1982 to May 31st 2021 and are presented in Table 2 for the SST component of the L4 and L3S datasets against drifting buoys, moored buoys and Argo floats. Separately, validation statistics for the temporal extension of 2024 are also included, using in situ observations from drifting buoys.

L4 and L3S SST (for points with SIC ≤ 15%) show almost zero biases compared to the corresponding in situ data, while standard deviations are around 0.5°C. The biases are lower than those reported in Castro et al. (2016), where nine different SST analyses and two single sensor satellite products were compared against independent observations from the Upper Temperature of the polar Oceans (UpTempO)[12] buoys deployed in the Beaufort Sea, north of Alaska in the Arctic Ocean, in 2012 and 2013. The standard deviations reported here are slightly lower than in Castro et al. (2016).

Table 2. Overall validation statistics from comparisons between the SST component of the Level 4 and Level 3S product and in situ observations. The table shows statistics for the period January 1st 1982 to May 31st 2021 and for the temporal extension of 2024. RMS: Root Mean Square; Nobs: number of observations.

Variable	**Unit**	**In Situ**	**Mean Bias**	**Standard deviation**	**RMS**	**Nobs**
Sea Surface Temperature SEAICE_ARC_PHY_CLIMATE _L4_MY_011_**016** 01/01/1982-31/05-2021	°C	Drifting buoys	0.00	0.54	0.54	3,062,549
		Moored buoys	0.03	0.56	0.56	76,052
		Argo	0.03	0.51	0.51	32,953
Sea Surface Temperature SEAICE_ARC_PHY_CLIMATE _L4_MY_011_**016** Temporal extension 2024	°C	Drifting buoys	-0.06	0.96	0.40	92452
Sea Surface Temperature SEAICE_ARC_PHY_CLIMATE_L3 S_MY_011_**021** 01/01/1982-31/05-2021	°C	Drifting buoys	0.07	0.43	0.44	876709
		Moored buoys	0.10	0.53	0.54	29946
		Argo	0.12	0.40	0.41	10312
Sea Surface Temperature SEAICE_ARC_PHY_CLIMATE_L3 S_MY_011_**021** Temporal extension 2024	°C	Drifting buoys	-0.12	0.94	0.33	30841

[12] See https://psc.apl.washington.edu/UpTempO/

For the temporal extension of 2024, results show similar mean differences as the previous validations, while the standard deviations are higher but still less than 1°C. This can be due to a short comparison period of June to October 2024 and thus, a lower number of in situ observations available.

Figures 4 and 5 show the yearly validation statistics for the L4 and L3S products, against drifting buoys, moored buoys and Argo floats. It should be noted that the amount of drifting and moored buoys are much lower in the beginning of the time period compared to the latest years, which makes the validation statistics against drifter and moorings less reliable in the beginning of the record. The Argo floats are only available from May 2001. Overall, the bias and standard deviation are both quite stable over the period of time for all types of in situ observations. The stability is particularly remarkable after the year 2000, when the in situ data availability increased.

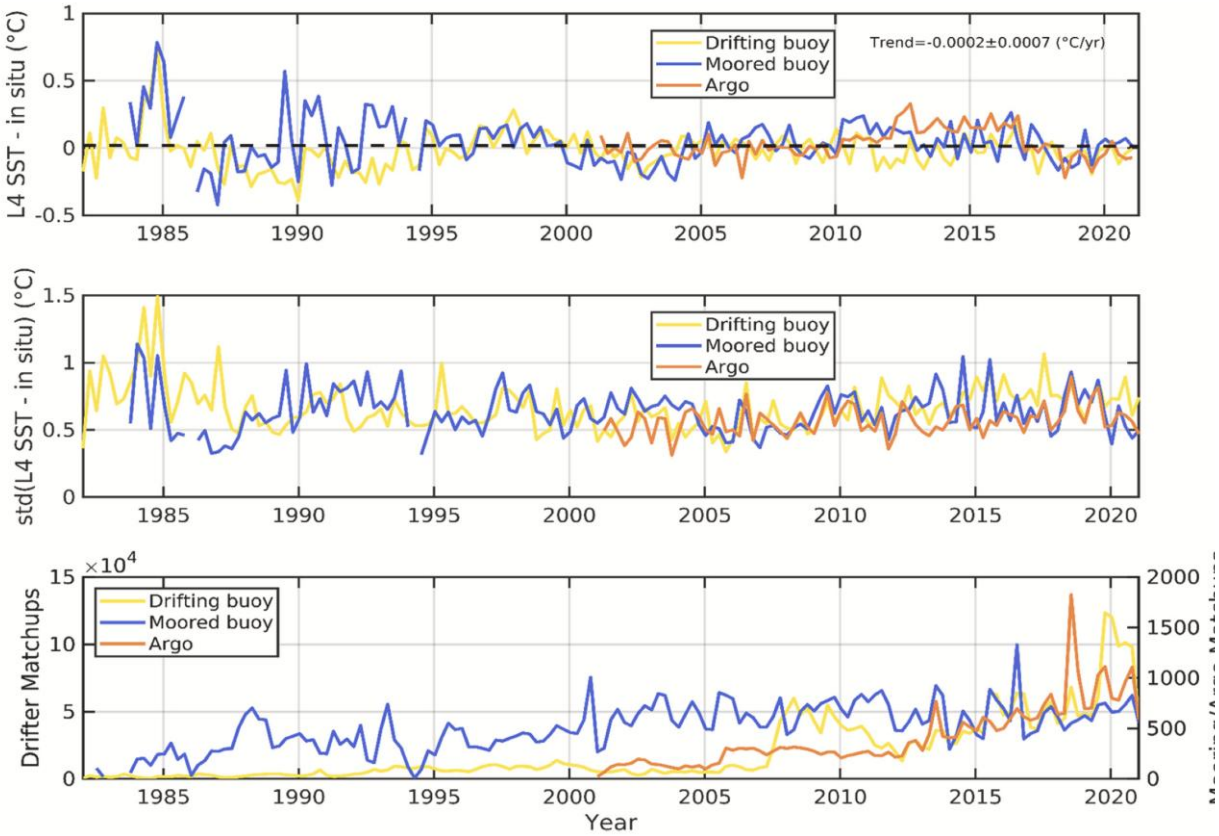

Figure 4. Seasonal SST validation statistics of the level 4 reanalysis (January 1st 1982 to May 31st 2021) from drifting buoys (yellow), moored buoys (blue) and Argo floats (orange). Mean differences (top), standard deviations (middle) and numbers of match-ups (bottom). The spatial and temporal coverage of the satellite analysis vs in situ match-ups throughout the record are shown in Figure 2.

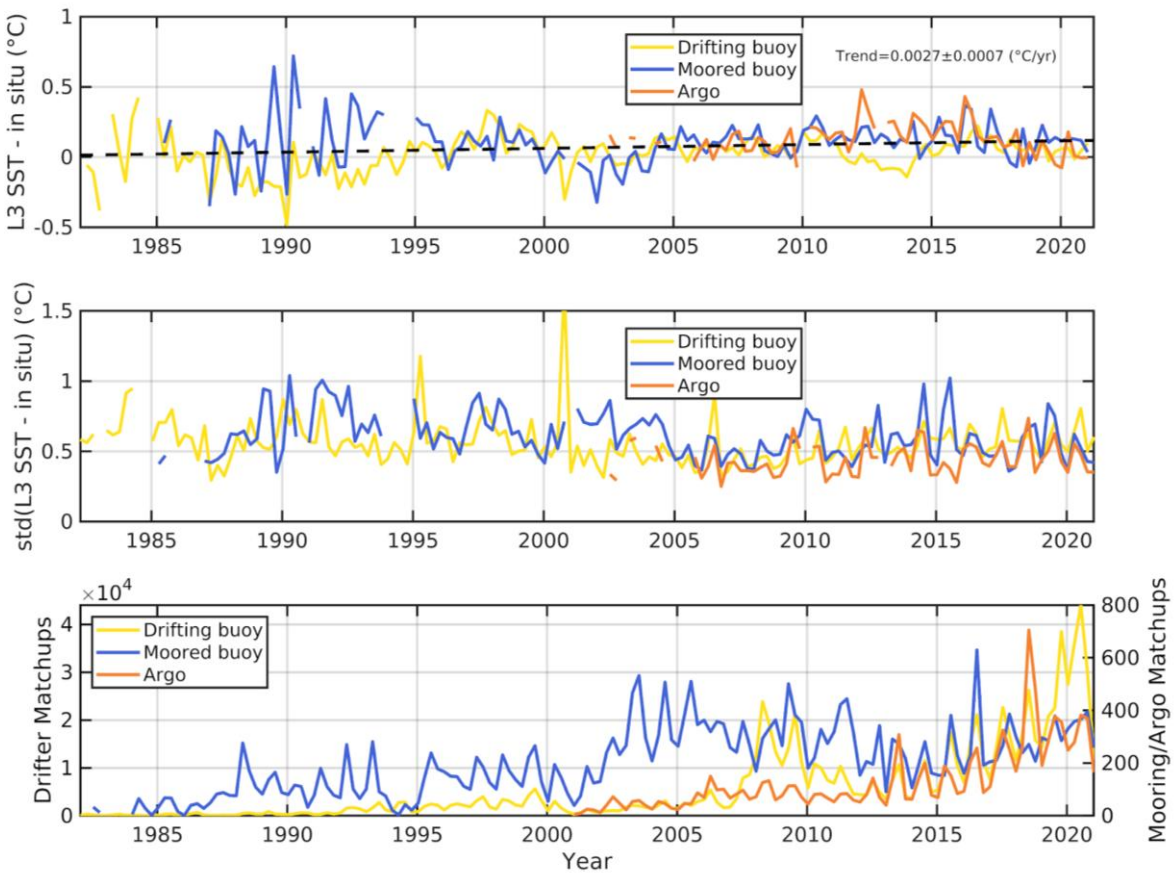

Figure 5. Seasonal SST validation statistics of the level 3S reanalysis (January 1st 1982 to May 31st 2021) from drifting buoys (yellow), moored buoys (blue) and Argo floats (orange). Mean differences (top), standard deviations (middle) and numbers of match-ups (bottom).

IV.2 Validation of IST

The L4 IST validation results cover multi-year periods for buoy data from ECMWF (September 15th 1993 to January 1st 2015) and CRREL (April 14th 2001 to September 4th 2017), NP drifting ice stations (1982-1989, 2003-2012) and IceBridge flight campaigns (during the period from March to May, 2012-2019) and are presented in Table 3. The L3S IST validation results cover multi-year periods using the same observations as above, except for the IceBridge flight campaigns (due to very few match-ups resulting from the gaps in the L3S product) and are also presented in Table 3. Furthermore, for the temporal extension of 2024, SIMB3 buoys were used to validate both the L4 and L3S datasets. These results are also shown in Table 3.

The validation of the L4 and L3S IST (SIC > 15%) is limited by the sparse number of in situ observations as well as increased in situ uncertainties in the ice-covered regions compared to the open ocean. The L4 and L3S IST are colder than in situ measurements from NP, ECMWF and CRREL by up to 3.2°C, while standard deviations range between 3.0°C and 3.3°C. NP, ECMWF and CRREL measure air temperatures at a varying height, typically of about 2 m (T_{2m}), instead of the surface temperature provided by the L4 and L3S products. Nielsen-Englyst et al. (2019) found an average IST-T_{2m} difference of -1.25°C during all-sky conditions over sea ice. This IST-T_{2m} difference is a real temperature difference between the snow surface and the air above it, which may partly explain the inflated mean differences observed in the

validation results, thus not related to the performance of the L4 IST. The L4 IST component is warmer by 1.5°C when compared to the IceBridge data, while the standard deviation of 3.1°C is similar to that of the comparisons with ECMWF, CRREL and NP in situ observations. Part of the bias against IceBridge may be explained by the missing cloud screening of IceBridge observations. If clouds are present between the Operation Icebridge Aircraft - that is used for the in situ data collection - and the surface, the radiometer will provide the temperature of the clouds (usually colder) instead of the surface temperature. Furthermore, IceBridge provides "snap-shots" of the surface temperature, while the L4 SST/IST provides daily means, which have been adjusted by using the 24 h average clear-sky bias correction.

Table 3. Overall validation statistics from comparisons between the IST component of the Level 4 and Level 3S reprocessed data set and in situ observations. The table shows statistics for the period January 1st 1982 to May 31st 2021 and for the temporal extension of 2024. RMS: Root Mean Square; Nobs: number of observations.

Variable	**Unit**	**In Situ**	**Mean Bias**	**Standard deviation**	**RMS**	**Nobs**
Ice Surface Temperature SEAICE_ARC_PHY_CLIMATE_L4_MY_011_**016** (01/01/1982-31/05-2021)	°C	ECMWF buoys (T2m)	-3.21	3.34	4.63	55,288
		CRREL buoys (T2m)	-2.87	3.36	4.42	22,979
		NP Drifting Ice Stations (T2m)	-2.35	3.12	3.91	7,665
		IceBridge KT-19	1.52	3.12	3.48	36,638
Ice Surface Temperature SEAICE_ARC_PHY_CLIMATE_L4_MY_011_**016** Temporal extension 2024	°C	SIMB3 (T2m)	-2.56	3.52	3.60	2862
Ice Surface Temperature SEAICE_ARC_PHY_CLIMATE_L3S_MY_011_**021** (01/01/1982-31/05-2021)	°C	ECMWF buoys (T2m)	-3.22	3.36	4.65	26502
		CRREL buoys (T2m)	-2.69	3.38	4.32	11149
		NP drifting ice stations (T2m)	-1.81	2.51	3.09	1183
Ice Surface Temperature SEAICE_ARC_PHY_CLIMATE_L3S_MY_011_**021** Temporal extension 2024	°C	SIMB3 (T2m)	-1.40	2.98	2.55	778

For the 2024 temporal extension, the validation with SIMB3 buoys shows very similar results to the previous validations against other drifters.

Annual mean biases and standard deviation for the L4 and L3S products against the in situ datasets are shown in Figures 6 and 7, clearly demonstrating the limited periods of the available in situ measurements. Overall, the behaviour of both the standard deviation and the bias is relatively stable

throughout the years. As of 2022, GCOS requirements exist for sea ice surface temperature, as IST is recognized as an Essential Climate Variable (ECV), see GCOS (2022). Note that the numbers given here are differences between satellite and in situ observations, and not errors in the satellite products. In addition, the use of air measurements to validate IST introduces a difference due to the vertical stratification in the near surface temperature, see Nielsen-Englyst et al. (2019).

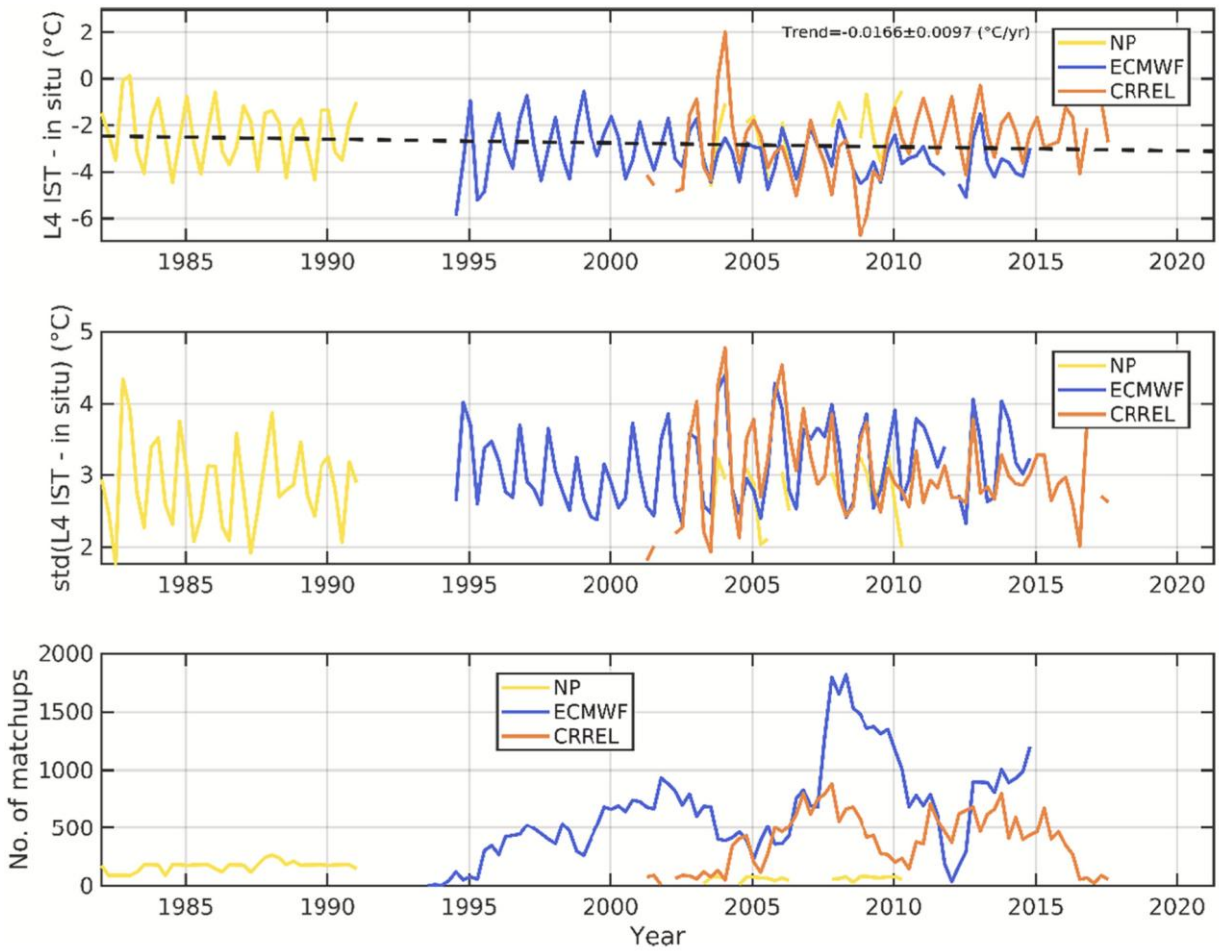

Figure 6. Seasonal IST validation statistics of the level 4 reanalysis (January 1st 1982 to May 31st 2021) from NP buoys (yellow), ECMWF buoys (blue) and CRREL buoys (orange). Mean biases (top), standard deviations (middle) and number of matchups (bottom). The spatial and temporal coverage of the satellite analysis vs in situ matchups throughout the record are shown in Figure 3.

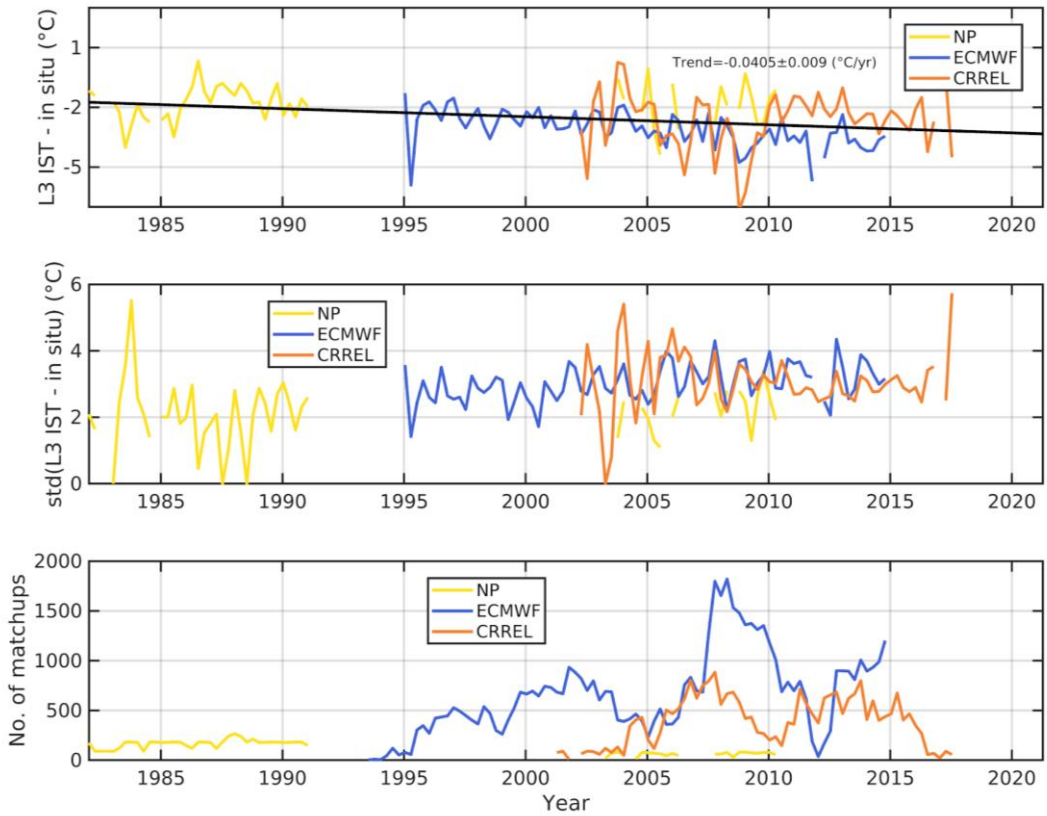

Figure 7. Seasonal IST validation statistics of the level 3S reanalysis (January 1st 1982 to May 31st 2021) from ECMWF buoys (blue), CRREL buoys (orange) and NP drifting stations (yellow). Mean biases are shown on the top panel, standard deviations in the middle panel and number of available match-ups in the bottom panel.

IV.3 Conclusion of the validation

The overall validation metrics for the SST component in both the L4 and the L3S datasets showed mean differences with drifting buoys well within the target requirements for satellite SST products (O'Carroll et al., 2019), especially for drifting buoys – which are considered reference sensors but also for Argo floats and moored buoys. For climate assessment purposes, the stability of the dataset was assessed by estimating the trend in the differences between the L4 SST and drifting buoys, available in Nielsen-Englyst et al. (2023), which was found to be 0.01°C per decade matching the GCOS requirement of 0.01 °C per decade for the goal stability and well within the target requirement of 0.1°C per decade. The threshold accuracy for SST, as defined by GCOS, is 0.3°C with a goal accuracy of 0.05°C, achieved for the SST component considering all types of in situ observations.

According to the most recent GCOS requirements for ECVs, the IST component has a 0.3°C per decade target stability while the goal stability is 0.1°C per decade. The threshold accuracy is 6°C while the goal accuracy is 1°C The L4 IST meets the threshold accuracy requirement in all cases of in situ observations used, although the comparison is with T2m, hence the large discrepancies reported. Regarding the stability, as reported by Nielsen-Englyst et al. (2023), this is met when the NP stations are used (which span the longest available period).

V SYSTEM'S NOTICEABLE EVENTS, OUTAGES OR CHANGES

This section is dedicated to describe and track changes that may occur in the current operational system as noted in Table 4. In includes changes due to outages (data gaps), upstream data (e.g., addition or loss of a sensor, inclusion/removal of in-situ platforms), reprocessing, corrections, assimilated data, boundary conditions, among other changes that may impact product quality.

Table 4. Description and tracking change records during the current reanalysis system.

Date	Change/Event description	System version	Other
8/09/2021	During the scheduled temporal extension, input data format was updated from L2P to L3U.	1.0	
31/10/2023	During the scheduled temporal extension from 2021 to 2022, a full reprocessing of the dataset occurred to account for erroneous issues of some upstream data.	2.0	

VI QUALITY CHANGES SINCE PREVIOUS VERSION

This section aims to provide a short summary of the main changes in Product Quality between the previous and current version.

VI.1 Regular time extension of L4 product

The L4 reprocessed product originally released covered the period January 1st 1982 to August 31st 2019. During the first temporal extension, the product was extended for the period September 1st 2019 to May 31st 2021, based on C3S SST L3U data instead of the L2P format. Further regular updates are based on C3S L3U instead of the L2P format. The main difference between the L2P and L3U product is in the spatial resolution of the L3U product, which is coarser (0.05 degrees) than the L2P product and the L4 product grid.

A consistency check was performed, using L2P and L3U C3S data for the same three-month period (June to August 2019) to create two separate L4 products. Figure 8 shows the mean difference for this test period, where 'L4_new' denotes the product created with the L3U input, while 'L4_old' is the previously created L4 product with the L2P input. The figure shows no large changes, with a total mean difference of 0.01°C, so the overall performance of the product is not affected by this change in input data format and spatial resolution.

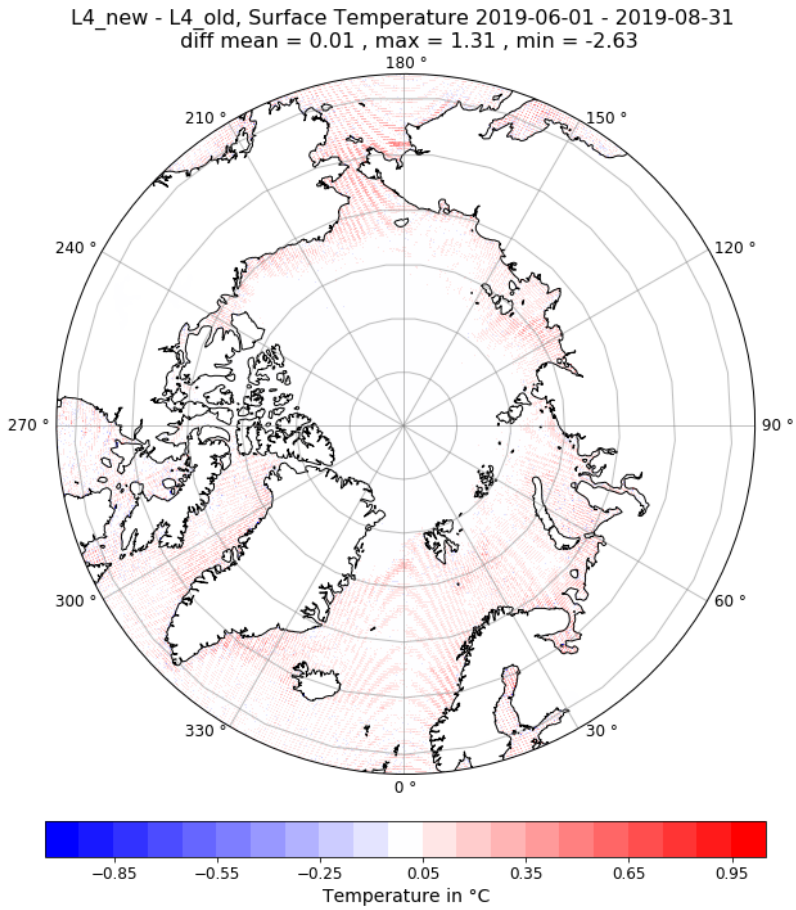

Figure 8. Mean difference between the L4 reanalysis created with L3U (L4_new) vs L2P (L4_old) input data from C3S for the 3-month period June to August 2019.

VI.2 Reprocessing of L4 product

In the reprocessing of the L4 product (from v1.0 to v2.0) several changes have been applied. These include 1) an update of the OI statistical parameters, 2) a change to using noise weighted average instead of prioritising input satellite data, 3) a small change in sea ice concentration filters, 4) a change in the upstream input SST data, 5) updated uncertainties for input satellite data and 6) updated bias correction schemes for post-processing. Only a few of these changes (i.e., No. 4 and 6) show significant impact on the accuracy and/or product content and these are changes are described in greater detail in the following.

Change no. 4) Change in the upstream input SST data: In v1.0, the AASTI SST was included in the OI processing over open ocean. However, an analysis has shown that the quality of the AASTI SST is inadequate and that significant improvements (i.e., lower SST standard deviations) are obtained by leaving out the AASTI observations over open ocean.

Change no. 6) Updated bias correction schemes for post-processing: In v1.0, a constant bias correction was applied everywhere based on the validation against drifters. In v2.0, a constant bias correction has also been applied based on the validation against drifters, but only over open ocean. In contrast, over sea ice, a constant clear-sky bias correction has been applied based on the findings in Nielsen-Englyst et al.(2019) to obtain all-sky daily averages. The clear-sky bias correction has been linearly scaled using the sea ice concentration as a scaling factor. This change results in warmer ISTs and resulting lower mean differences against NP, ECMWF and CRREL buoys, but a larger average mean difference against IceBridge.

A summary of changes in the validation statistics by comparing the old statistics (v1.0) to those obtained from the new reprocessed L4 product (v2.0) is presented in Table 5, showing improved statistics for v2.0 compared to v1.0, in the form of lower mean bias for all in situ observations considered, for both SST and IST. The SST component experience lower standard deviation and Root mean square for V2, while for the IST component the standard deviation remains unchanged and root mean square lower for three out of four of the in situ sources. Pearson correlation coefficient values are almost unchanged.

Table 5. Overall L4 validation statistics against in situ observations for comparison of v1.0 and v2.0. The table shows statistics for the period January 1st 1982 to May 31st 2021. RMS: Root Mean Square; Corr: correlation coefficient; Nobs: number of observations.

In Situ	Unit	Mean Bias		Standard deviation		RMS		Corr		Nobs	
		V1	V2	V1	V2	V1	V2	V1	V2	V1	V2
Drifting buoys SST	°C	0.04	0.00	0.62	0.54	0.62	0.54	0.98	0.99	3,075,054	3,062,549
Moored buoys SST		0.08	0.03	0.61	0.56	0.62	0.56	0.98	0.98	76,043	76,052
Argo SST		0.08	0.03	0.60	0.51	0.60	0.51	0.99	0.99	33,043	32,953
NP Drifting Ice Stations (T2m) IST		-2.88	−2.35	3.12	3.12	4.24	3.91	0.98	0.98	7660	7665
ECMWF buoys (T2m) IST		-3.74	−3.21	3.34	3.34	5.01	4.63	0.96	0.96	55286	55288
CRREL buoys (T2m) IST		-3.40	−2.87	3.36	3.36	4.78	4.42	0.96	0.96	22959	22979
IceBridge KT-19 IST		0.98	1.52	3.13	3.12	3.27	3.48	0.92	0.92	36638	36638

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